

```
115 .model MS1 SW(Ron={Ron} Roff=1e9 Vt=0.9 Vh=0 Lser=0 Vser=0)
116 E1 phix 0 value={psi_0}
117 .ENDS ;
118
119 * magnetic leg
120 .SUBCKT Leg_Yy m1 m2
121 +PARAMS: N=1 RFe=1 ImyN=1 a=1 phiN=1 m=9 ctrl=1
122 LRFem1 mh {N*ctrl*ImyN/RFe} Rser={0.001m*N*N/RFe} ; Rser=parasitic Value
123 vphi mh mh1 0
124 rphi mh 0 1e12
125 Eimymh1 m2
126 +VALUE={N*ImyN*(a*i(vphi)/phiN+(1-a)*PWRS(i(vphi)/phiN,m))}
127 .ENDS;
```